

Dengue Surveillance Analysis Dashboard

Global Epidemiological Analysis of Dengue Cases, Severity, and Mortality

Prepared by: Oumar Mahamat Ahmat

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Executive Summary

Dengue is a major global public health concern, particularly in tropical and subtropical regions. This project analyzes global dengue surveillance data to examine patterns in disease burden, severity, and mortality across countries and WHO regions.

The analysis followed a structured epidemiological data workflow, including data quality assessment, cleaning, feature engineering, exploratory data analysis, statistical analysis, and interactive visualization using Power BI.

Key indicators included reported dengue cases, confirmed cases, severe cases, deaths, severe case rate, and case fatality rate (CFR). Geographic and temporal analyses were used to identify areas with high disease burden, elevated severity, and increased mortality.

The resulting Dengue Surveillance Analysis Dashboard provides an interactive tool for exploring dengue patterns by time, country, and WHO region. The findings can support surveillance activities, outbreak detection, resource prioritization, and public health decision-making.

Background

Dengue is a viral infection transmitted primarily by *Aedes aegypti* and *Aedes albopictus* mosquitoes. Dengue infection can range from asymptomatic or mild illness to severe dengue, which may result in plasma leakage, severe bleeding, organ impairment, shock, and death.

The epidemiology of dengue is influenced by multiple factors, including climate, urbanization, population density, mosquito distribution, environmental conditions, human mobility, and the effectiveness of vector-control programs.

Routine surveillance data provide an important foundation for monitoring these patterns. However, large surveillance datasets can be difficult to interpret without appropriate data visualization and epidemiological indicators.

This project therefore applies data analytics and visualization techniques to convert dengue surveillance data into an accessible decision-support dashboard.

Project Objectives

1/ General Objective

To develop an interactive epidemiological dashboard for analyzing the global distribution, severity, and mortality of dengue disease.

2/ Specific Objectives

The project aims to:

- 1) Quantify the overall number of reported dengue cases.
- 2) Identify countries and WHO regions with the highest dengue burden.
- 3) Analyze the distribution of severe dengue cases.
- 4) Estimate severe case rates across geographic areas.
- 5) Examine dengue-associated deaths.
- 6) Calculate and compare case-fatality rates (CFR).
- 7) Identify temporal and geographic patterns in dengue surveillance.
- 8) Provide an interactive platform for exploring epidemiological indicators.
- 9) Support evidence-based interpretation of dengue surveillance data.
- 10) Demonstrate the application of epidemiological and business intelligence techniques using Power BI.

Data and Methodology

1/ Data Source

The project uses a dengue surveillance dataset containing epidemiological information at country and regional levels.

The dataset includes variables related to:

- Country
- Geographic region
- Year

- Total dengue cases
- Confirmed cases
- Severe cases
- Deaths
- Population and/or population-related indicators
- Case-fatality measures

The dataset was prepared and analyzed for visualization in Microsoft Power BI.

2/ Data Preparation

Before visualization, the dataset underwent a data preparation process that included:

- Reviewing variable names and data types.
- Identifying missing values.
- Checking for duplicate records.
- Standardizing geographic names.
- Ensuring numerical variables were correctly formatted.
- Checking the consistency of epidemiological indicators.
- Creating calculated measures required for dashboard analysis.

Data preparation was important to ensure that the dashboard produced consistent and interpretable results.

Epidemiological Indicators

Several key indicators were incorporated into the dashboard.

1/ Total Cases

Total reported dengue cases provide an overall measure of disease burden.

Total Cases = Sum of reported dengue cases

This indicator is useful for identifying countries or regions experiencing the largest absolute number of reported cases.

2/ Severe Cases

Severe cases represent dengue infections associated with severe clinical outcomes.

Severe Cases = Sum of reported severe dengue cases

This indicator helps identify areas where dengue may be producing a greater burden of serious disease.

3/ Severe Case Rate

The severe case rate measures the proportion of reported dengue cases classified as severe.

Severe Case Rate (%) = Severe Cases / Total Cases × 100

This measure allows comparison between areas with different numbers of total cases.

4/ Deaths

Deaths represent reported dengue-associated fatalities.

Deaths = Sum of reported dengue deaths

The indicator provides an important measure of the most serious outcome of dengue infection.

5/ Case-Fatality Rate

Case-fatality rate measures the proportion of reported dengue cases that resulted in death.

CFR (%) = Deaths / Total Cases × 100

CFR is particularly useful for comparing mortality outcomes between geographic areas or time periods.

However, CFR should be interpreted with caution because it can be affected by differences in case detection, reporting completeness, healthcare access, diagnostic practices, and the clinical severity of detected cases.

Dashboard Design

The dashboard was developed as an interactive Power BI application consisting of multiple analytical pages.

1/ Global Overview

The Global Overview page provides a high-level summary of the dengue surveillance situation.

The page contains key performance indicators including:

- Total Cases
- Confirmed Cases
- Severe Cases
- Deaths
- CFR

These KPI cards allow users to quickly understand the overall magnitude of dengue.

Geographic Analysis

The Geographic Analysis page focuses on the spatial distribution of dengue.

Geographic visualization is particularly important in epidemiology because dengue transmission is highly heterogeneous.

The analysis allows users to compare:

- Countries
- WHO regions
- Total cases
- Severe cases
- Deaths
- Case-fatality rates

A geographic visualization can help identify areas with high absolute disease burden, while regional comparisons provide a broader understanding of disease distribution.

Importantly, a country with the largest number of cases is not necessarily the country with the highest incidence or mortality risk. Therefore, absolute counts should be interpreted alongside population-adjusted and outcome-based indicators where available.

Severity and Mortality Analysis

The Severity & Mortality page examines the clinical consequences of dengue.

The main indicators include:

- Severe Cases
- Severe Case Rate
- Deaths
- Case-Fatality Rate
- Severe Cases by Region

Comparing severe cases by WHO region helps identify geographic areas contributing substantially to the global burden of severe dengue.

❖ Severe Case Rate

The severe case rate provides additional context by measuring severe cases relative to total reported cases.

❖ Mortality

Deaths provide an absolute measure of the fatal burden of dengue.

❖ Case-Fatality Rate

CFR complements the mortality count by describing the proportion of reported cases resulting in death.

Together, these indicators provide a more comprehensive assessment than using case counts alone.

Epidemiological Interpretation

The dashboard demonstrates several important principles of epidemiological analysis.

1/ Disease Burden Is Not Uniform

Dengue transmission varies considerably across geographic areas. Some regions may contribute a large proportion of global cases because of population size and sustained transmission.

2/ High Case Counts Do Not Necessarily Mean High Mortality

A region can report many dengue cases while maintaining a relatively low CFR. Conversely, a region with fewer reported cases may have a higher CFR.

This difference emphasizes the importance of analyzing both disease burden and disease outcomes.

3/ Severity Provides Additional Epidemiological Information

Total cases alone do not describe the complete health impact of dengue.

Monitoring severe cases and the severe case rate helps identify areas where dengue may be producing substantial clinical pressure.

4/ Mortality Requires Context

Reported deaths should be interpreted together with total cases, healthcare access, surveillance quality, diagnostic capacity, and reporting completeness.

A high CFR may reflect genuinely poor outcomes, but it can also occur when mild cases are underdiagnosed or when surveillance systems capture mainly severe cases.

Public Health Implications

The findings generated through the dashboard can support several public health activities.

1/ Surveillance

Continuous monitoring of cases, severe disease, and deaths can help surveillance teams identify changes in dengue patterns.

2/ Early Warning

Rapid increases in reported cases or severe cases may provide an indication of emerging outbreaks and the need for intensified investigation.

3/ Resource Allocation

Geographic identification of high-burden areas can support prioritization of:

- Healthcare resources
- Diagnostic capacity
- Surveillance activities
- Vector-control interventions
- Community awareness campaigns
- Clinical preparedness

4/ Clinical Preparedness

Monitoring severe cases and deaths can help health systems anticipate periods of increased demand for emergency and inpatient care.

5/ Vector Control

Geographic and temporal patterns identified through surveillance can contribute to targeted mosquito-control strategies.

Limitations

Several limitations should be considered when interpreting the dashboard.

1/ Reporting Bias

Surveillance data may not represent all dengue infections because many mild or asymptomatic infections may never be diagnosed or reported.

2/ Differences in Surveillance Systems

Countries may differ in diagnostic capacity, reporting practices, case definitions, and surveillance completeness.

3/ Missing Data

Missing observations can affect comparisons between countries, regions, and years.

4/ CFR Interpretation

CFR is affected by case ascertainment. If mild cases are underreported, CFR may be artificially elevated.

5/ Ecological Interpretation

Country-level or regional patterns should not automatically be interpreted as individual-level risk factors.

6/ Temporal Comparability

Changes in surveillance systems, diagnostic practices, reporting policies, or healthcare access over time may influence observed trends.

Recommendations

Based on the analytical framework, the following recommendations are proposed:

1. Strengthen routine dengue surveillance to improve the completeness and timeliness of reporting.
2. Monitor severe cases and deaths alongside total cases rather than relying on case counts alone.
3. Prioritize high-burden geographic areas for enhanced surveillance and vector-control activities.
4. Investigate areas with unusually high CFRs to determine whether the pattern reflects genuine mortality risk or differences in case detection.
5. Improve laboratory and diagnostic capacity to strengthen case confirmation.

6. Integrate epidemiological, environmental, and climatic data to improve dengue early-warning systems.
7. Use population-adjusted indicators, such as incidence rates, when comparing countries with substantially different population sizes.
8. Maintain interactive dashboards to support real-time or near-real-time surveillance decision-making.
9. Strengthen data quality assurance through standardized definitions, validation procedures, and regular reporting reviews.

Conclusion

The Dengue Surveillance Analysis Dashboard demonstrates how epidemiological data can be transformed into an interactive decision-support tool.

By combining total cases, confirmed cases, severe cases, deaths, severe case rates, and case-fatality rates with geographic analysis, the dashboard provides a multidimensional view of dengue disease burden and outcomes.

The project highlights an important principle of epidemiological surveillance: no single indicator is sufficient to describe the impact of a disease. Case counts describe the magnitude of disease, severity indicators describe the clinical burden, and mortality indicators provide information about the most serious outcomes.

Power BI provides an effective platform for integrating these indicators into an accessible and interactive surveillance product. When combined with sound epidemiological interpretation and appropriate data-quality assessment, such dashboards can support surveillance, resource allocation, outbreak preparedness, and public health decision-making.

Overall, this project demonstrates the integration of epidemiology, health data analytics, data visualization, and business intelligence to address a real-world public health problem.

Technical Tools

The project utilized the following tools and analytical approaches:

Component	Tool/Approach
Data Preparation	Microsoft Excel / Power Query
Data Analysis	Epidemiological calculations
Data Visualization	Microsoft Power BI
Dashboard Development	Power BI
Geographic Analysis	Maps and regional visualizations
Indicators	Cases, severe cases, deaths, severe case rate, CFR
Data Storytelling	Interactive KPI cards and charts

Project Deliverable

The final product is an interactive **Dengue Surveillance Analysis Dashboard** designed to communicate epidemiological findings clearly to both technical and non-technical audiences.

The dashboard can serve as a portfolio project demonstrating practical competencies in:

- Epidemiological analysis
- Health data analytics
- Surveillance data interpretation
- Data visualization
- Power BI
- KPI development
- Geographic analysis
- Mortality analysis
- Data storytelling

- Public health decision support

Project Title: Dengue Surveillance Analysis Dashboard

Subtitle: Global Epidemiological Analysis of Dengue Cases, Severity, and Mortality

Primary Tool: Microsoft Power BI

Domain: Epidemiology & Public Health Data Analytics